Student Performance Prediction Using Machine Learning

Abstract

This project uses machine learning techniques to predict student performance based on academic, demographic, and lifestyle factors, achieving high accuracy using classification models.

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Student Performance Prediction Using Machine Learning

1. Introduction

Education plays a crucial role in shaping an individual's future, and predicting student performance can help educators take proactive measures to improve outcomes. This project focuses on applying machine learning techniques to predict whether a student will pass or fail based on various academic, demographic, and lifestyle factors.

The model analyzes patterns in student data and provides insights into the key factors influencing academic success.

2. Objectives

The main objectives of this project are:

- To build a predictive model for student performance
- To perform data preprocessing and transformation
- To apply machine learning classification algorithms
- To compare model performance
- To identify the most important factors affecting student results

3. Dataset Description

- **Dataset Name:** Student Performance Dataset (Mathematics)
- **File Used:** student-mat.csv
- **Source:** UCI Machine Learning Repository
- **Dataset Link:** https://archive.ics.uci.edu/ml/datasets/student+performance

Dataset Details:

- Total Records: **395 students**
- Total Features: **33 variables**
- Format: CSV (semicolon-separated)

Key Features:

- **Demographics:** age, gender, address
- **Family Background:** parental education (Medu, Fedu), jobs
- **Academic Factors:** study time, failures
- **Lifestyle:** alcohol consumption (Dalc, Walc), social activities (goout)
- **Grades:**
    - G1 → First period grade

- G2 → Second period grade
- G3 → Final grade

4. Data Preprocessing

✓ Missing Values

The dataset was checked for missing values using `.isnull()` and no missing values were found. However, a preprocessing step using `SimpleImputer` was included for robustness.

✓ Encoding Categorical Data

Categorical variables such as gender, school, and job were converted into numerical form using **Label Encoding**.

✓ Feature Engineering

A new target variable `result` was created:

- **1 (Pass):** if G3 ≥ 10
- **0 (Fail):** if G3 < 10

5. Data Splitting

The dataset was divided into:

- **Training Set:** 80%
- **Testing Set:** 20%

This ensures that the model is evaluated on unseen data.

6. Feature Scaling

Feature scaling was performed using **StandardScaler** to normalize the data. This improves the performance of machine learning models, especially Logistic Regression.

7. Machine Learning Models Used

◆ Logistic Regression

- A linear classification algorithm
- Suitable for binary classification problems
- Used as a baseline model

◆ **Random Forest Classifier**

- An ensemble learning method
- Combines multiple decision trees
- Provides higher accuracy and feature importance

8. Results & Outputs

✓ **Model Accuracy**

- **Logistic Regression Accuracy:** 91.13%
- **Random Forest Accuracy:** 97.46%

☞ Random Forest outperformed Logistic Regression significantly.

✓ **Classification Report (Random Forest)**

Metric	Class 0 (Fail)	Class 1 (Pass)
Precision	0.96	1.00
Recall	1.00	0.94
F1-Score	0.98	0.97

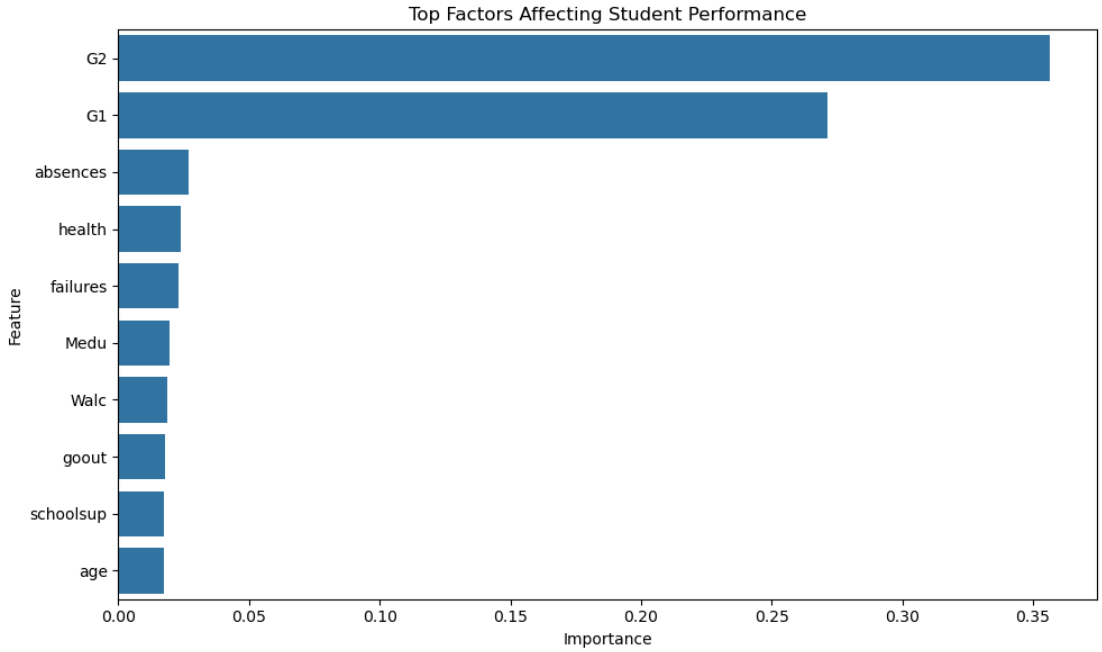
- **Overall Accuracy:** 97%
- The model shows balanced performance with high precision and recall.

9. Feature Importance Analysis

Top features influencing student performance:

Feature	Importance
G2	35.66%
G1	27.13%
Absences	2.68%
Health	2.39%
Failures	2.29%
Medu	1.97%
Walc	1.87%
Goout	1.78%
Schoolsup	1.74%
Age	1.74%

10. Graph Analysis



The bar chart titled **"Top Factors Affecting Student Performance"** visually represents feature importance.

Observations:

- **G2 (second period grade)** is the most important factor
- **G1 (first period grade)** is the second most important
- Both dominate the graph significantly

Additional Factors:

- Absences and failures negatively impact performance
- Health and lifestyle factors have moderate influence
- Family background (Medu) contributes slightly

11. Key Insights

- Students with higher previous grades are more likely to pass
- Attendance (absences) plays an important role
- Health and lifestyle habits influence outcomes
- Family education level has moderate impact

13. Future Improvements

- Remove G1 and G2 for more realistic prediction
- Apply hyperparameter tuning (GridSearchCV)
- Use advanced models like XGBoost and SVM
- Deploy the model as a web application using Streamlit

14. Conclusion

This project successfully applied machine learning techniques to predict student performance. The Random Forest model achieved the highest accuracy (**97.46%**). Feature importance analysis revealed that previous academic performance is the strongest predictor, followed by attendance, health, and lifestyle factors.